MATH 152
Mrs. Bonny Tighe

QUIZ 3A
7.2, 7.3
25 points

Name Solution

TA_____ Wed 2/21/2024

1. Evaluate the integrals.

a) $\int \tan^2 x \sec^4 x\, dx$

$\int \tan^2 x (\sec^2 x) \sec^2 x\, dx$

$\int \tan^2 x (1+\tan^2 x) \sec^2 x\, dx$

$u = \tan x$
$du = \sec^2 x\, dx$

$\int (\tan^2 x + \tan^4 x) \sec^2 x\, dx$

$\frac{1}{3}\tan^3 x + \frac{1}{5}\tan^5 x + C$

b) $\int t\,(\sin^2 t)\, dt$

$u = t$
$du = dt$

$dv = \sin^2 t = \frac{1}{2} - \frac{1}{2}\cos 2t\, dt$
$v = \frac{1}{2}t - \frac{1}{4}\sin 2t$

$t(\frac{1}{2}t - \frac{1}{4}\sin 2t) - \int (\frac{1}{2}t - \frac{1}{4}\sin 2t)\, dt$

$\frac{1}{2}t^2 - \frac{t}{4}\sin 2t - \frac{1}{4}t^2 - \frac{1}{8}\cos 2t + C$

c) $\int \tan^3(2x)\, dx$ $= \int \tan(2x) \cdot \tan^2(2x)\, dx$

$\int \tan(2x)(\sec^2(2x) - 1)\, dx =$

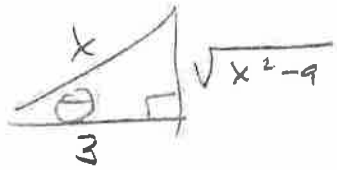
$\int \tan(2x) \sec^2 2x\, dx - \int \tan(2x)\, dx$

$\frac{1}{2} \cdot \frac{1}{2}\tan^2(2x) - \frac{1}{2}\ln|\sec 2x| + C$

d) $\int \frac{\cancel{x^2}\, x^3}{\sqrt{x^2-9}}dx$

$x = 3\sec\theta \qquad \sec\theta = x/3$

$dx = 3\sec\theta\tan\theta\, d\theta$



$$\int \frac{(3\sec\theta)^3\, 3\sec\theta\tan\theta\, d\theta}{\sqrt{9\sec^2\theta-9}} = \int \frac{27\sec^4\theta\,\cancel{\tan\theta}}{\cancel{\tan\theta}}\, d\theta$$

$$27\int \sec^2\theta(1+\tan^2\theta)\,d\theta = 27\int \sec^2\theta\, d\theta + 27\int \tan^2\theta\sec^2\theta\, d\theta$$

$$27\tan\theta - 27\cdot\tfrac{1}{3}\tan^3\theta + C = 27\left(\frac{\sqrt{x^2-9}}{3}\right) - 9\left(\frac{\sqrt{x^2-9}}{3}\right)^3 + C$$

e) $\int_2^{\sqrt{3}} \frac{\sqrt{4-x^2}}{x^2}dx$

$x = 2\sin\theta \qquad \sqrt{3} = 2\sin\theta \quad \theta = \pi/3$

$dx = 2\cos\theta\, d\theta \qquad 2 = 2\sin\theta \quad \theta = \pi/2$

$$\int_{\pi/3}^{\pi/2} \frac{\sqrt{4-4\sin^2\theta}}{(2\sin\theta)^2}\, 2\cos\theta\, d\theta = \int_{\pi/3}^{\pi/2} \frac{4\cos^2\theta}{4\sin^2\theta}\, d\theta$$

$$\int_{\pi/3}^{\pi/2} \cot^2\theta\, d\theta = \int_{\pi/3}^{\pi/2} (\csc^2\theta - 1)\, d\theta = -\cot\theta - \theta \Big|_0^{\pi/2}$$

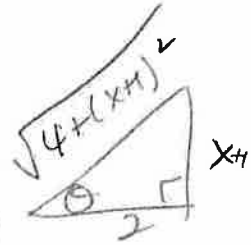
$$(-\cot\pi/2 - \pi/2) - (-\cot\pi/3 - \pi/3)$$

$$-0 - \pi/2 + \frac{\sqrt{3}}{3} + \pi/3 = \frac{\sqrt{3}}{3} - \pi/6$$

f) $\int \frac{1}{\sqrt{x^2+2x+5}}dx \qquad (x+1)^2+4$

$(x+1) = 2\tan\theta$

$dx = 2\sec^2\theta\, d\theta$



$$\int \frac{1}{\sqrt{(x+1)^2+4}}dx = \int \frac{1}{\sqrt{4\tan^2\theta+4}}(2\sec^2\theta\, d\theta)$$

$$\int \frac{2\sec^2\theta\, d\theta}{2\sec\theta} = \int \sec\theta\, d\theta =$$

$$\ln|\sec\theta + \tan\theta| + C = \ln\left|\frac{\sqrt{(x+1)^2+4}}{2} + \frac{x}{2}\right| + C$$

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Name Solutions

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1. Evaluate the integrals.

a) $\int t(\cos^2 t)\, dt$

$u = t \qquad dv = \cos^2 t = \frac{1}{2} + \frac{1}{2}\cos 2t\, dt$

$du = dt \qquad v = \frac{1}{2}t + \frac{1}{4}\sin 2t$

$t\left(\frac{1}{2}t + \frac{1}{4}\sin 2t\right) - \int\left(\frac{1}{2}t + \frac{1}{4}\sin 2t\right) dt$

$\frac{1}{2}t^2 + \frac{t}{4}\sin 2t - \frac{1}{4}t^2 - \frac{1}{8}\cos 2t + C$

b) $\int \sin^3(3x)\, dx = \int \sin 3x \sin^2(3x)\, dx = \int \sin 3x\,(1 - \cos^2 3x)\, dx$

$\int \sin 3x\, dx + \int \cos^2(3x)(-\sin 3x)\, dx$

$-\frac{1}{3}\cos 3x + \frac{1}{3}\cdot\frac{1}{3}\cos^3(3x) + C$

$-\frac{1}{3}\cos(3x) + \frac{1}{9}\cos^3(3x) + C$

c) $\int \tan^3 x \sec^2 x\, dx = \int \tan^2 x \sec x \sec x \tan x\, dx$

$\int (\sec^2 x - 1)\sec x \sec x \tan x\, dx$

$\int (\sec^3 x - \sec x)\sec x \tan x\, dx$

$u = \sec x$

$du = \sec x \tan x\, dx$

$\frac{1}{4}\sec^4 x - \frac{1}{2}\sec^2 x + C$

f) $\int_{3\sqrt{2}}^{6} \frac{x^3}{\sqrt{x^2-9}}dx$

$x = 3\sec\theta$
$dx = 3\sec\theta\tan\theta\, d\theta$

$6 = 3\sec\theta$
$2 = \sec\theta$
$\pi/3 = \theta$

$3\sqrt{2} = 3\sec\theta$
$\sqrt{2} = \sec\theta$
$\pi/4 = \theta$

$$\int_{\pi/4}^{\pi/3} \frac{27\sec^3\theta}{\sqrt{9\sec^2\theta - 9}}(3\sec\theta\tan\theta\, d\theta)$$

$$27\int_{\pi/4}^{\pi/3} \sec^4\theta\, d\theta = 27\int_{\pi/4}^{\pi/3} \sec^2\theta(\sec^2\theta)\, d\theta = \int \sec^2\theta(1+\tan^2\theta)\, d\theta$$

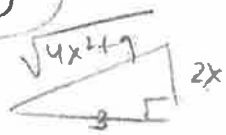
$$27\int_{\pi/4}^{\pi/3} \sec^2\theta\, d\theta + \int \tan^2\theta\sec^2\theta\, d\theta = 27\left[\tan\theta + \frac{1}{3}\tan^3\theta\right]_{\pi/4}^{\pi/3}$$

$$27\left[\left(\tan\pi/3 + \frac{1}{3}(\tan\pi/3)^3\right) - \left(\tan\pi/4 + \frac{1}{3}\tan^3\pi/4\right)\right] = 27\left[\sqrt{3} + \frac{1}{3}(\sqrt{3})^3 - (1 + 1/3)\right]$$

$$27\left[\sqrt{3} + \sqrt{3} - 4/3\right] = 27(2\sqrt{3} - 4/3)$$

h) $\int \frac{1}{\sqrt{4x^2+9}}dx$

$2x = 3\tan\theta$
$2x/3 = \tan\theta$
$x = 3/2\tan\theta \quad dx = 3/2\sec^2\theta\, d\theta$



$$\int \frac{1}{\sqrt{9\tan^2\theta + 9}}(3/2\sec^2\theta)\, d\theta$$

$$\int \frac{3/2\sec^2\theta}{3\sec\theta}d\theta = \frac{1}{2}\int \sec\theta\, d\theta$$

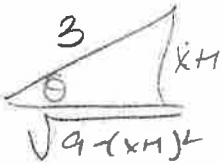
$$\frac{1}{2}\ln|\sec\theta + \tan\theta| + C$$

$$\frac{1}{2}\ln\left|\frac{\sqrt{4x^2+9}}{3} + \frac{2x}{3}\right| + C$$

i) $\int \sqrt{8 - x^2 - 2x}\, dx$

$-(x^2 + 2x + 1) + 8 + 1$
$-(x+1)^2 + 9$

$x + 1 = 3\sin\theta$
$dx = 3\cos\theta\, d\theta$



$$\int \sqrt{9 - (x+1)^2}\, dx$$

$$\int \sqrt{9 - 9\sin^2\theta}\, 3\cos\theta\, d\theta = 9\int \cos^2\theta\, d\theta$$

$$9\int\left(\frac{1}{2} + \frac{1}{2}\cos 2\theta\right)d\theta = \frac{9}{2}\theta + \frac{9}{4}\sin 2\theta + C$$

$$\frac{9}{2}\theta + \frac{9}{2}\sin\theta\cos\theta + C = \frac{9}{2}\sin^{-1}\left(\frac{x+1}{3}\right) + \frac{9}{2}\left(\frac{x+1}{3}\right)\left(\frac{\sqrt{9-(x+1)^2}}{3}\right) + C$$